

LITHIUM ION CONDUCTORS IN THE SYSTEM $AB(IV)_2(PO_4)_3$ ($B = Ti, Zr$ AND Hf)

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Lithium ion conductors based on $LiB_2(PO_4)_3$ ($B = Ti, Zr$ and Hf) type structures have been synthesized by suitable substitution at the B site. Solid solutions of the type $LiZr_{2-x}Ti_x(PO_4)_3$, $Li_{1+x}Sc_xTi_{2-x}(PO_4)_3$ and $Li_{1+x}In_xHf_{2-x}(PO_4)_3$ are found to exist for various values of x and their lithium conductivities are determined. The results are discussed in terms of the number of conductors, bottleneck size and stoichiometry.

1. INTRODUCTION

Lithium ion conductors are especially attractive for a number of reasons. Lithium is relatively inexpensive, has an extremely high reduction potential ($E_{red}^0 = -3.024V$) and is light in weight. Thus, lithium based batteries would have a high open circuit voltage and exhibit high energy density.¹ Therefore, a good lithium conductor, with the requisite ceramic properties, has been a goal of our research for a number of years. In this paper we concern ourselves with lithium ion conductors, based upon the $LiM(IV)_2(PO_4)_3$ framework. These compounds are related to the NASICON solid solutions of composition $Na_{1+x}Zr_2Si_xP_{3-x}O_{12}$. The end member, $NaZr_2(PO_4)_3$, has a low conductivity, but as x increases so does the conductivity reaching a maximum for $1.8 \leq x \leq 2.3$.² A silicate containing lithium ion series could not be prepared, but $Li_3Zr_2Si_2PO_{12}$ was obtained from the corresponding sodium analogue by ion exchange.²

Its conductivity is three orders of magnitude lower than the conductivity of its sodium ion counterpart. A search by Hong for additional framework structures which would satisfy the criteria³ for fast Li^+ conduction led to the discovery of $Li_{14}Zn(GeO_4)_4$. However, the $M(IV)_2(PO_4)_3$ framework allows a large number of substitutions to be made with retention of

basic structure and properties.⁵⁻⁹ Thus, it was felt that fast lithium ion conductors could be prepared by a combination of nonstoichiometry and proper framework substitutions. This paper details the results of our initial efforts.¹⁰

2. EXPERIMENTAL SECTION

2.1 Synthetic Procedure

Li_3PO_4 , TiO_2 and ZrO_2 (hafnium free) were spec. pure grade from Johnson Matthey, Inc. ZrP_2O_7 and TiP_2O_7 were prepared by heating α -zirconium phosphate¹¹ and α -titanium phosphate¹² respectively, at 1000°C for 4 hr. The phosphates in turn were prepared by standard procedures.^{11,12} The pyrophosphates were dissolved in dilute HF, analyzed by ICP and found to contain less than 100ppm of impurities. Sc_2O_3 , In_2O_3 and HfO_2 from Aldrich Chemical Co. were 99.99% pure.

The reagents (ZrO_2 , TiO_2 , Li_3PO_4 and Sc_2O_3) were dried in an oven at 300°C for 4-8 hr just prior to use and stored in a desiccator. Exact mole ratios of the ingredients were weighed out and ground together in an agate mortar until thoroughly blended. The entire solid was then transferred to a platinum tube (4mm O.D.) and sealed by cold pressing. The tube was heated at 800°C for 6 hr and then for 20 hr in the temperature range 1000-1100°C.

Any tube which had expanded due to gas evolution or which had lost more than 0.5% weight was discarded.

2.2. Instrumental Procedures

Powder patterns were recorded with a Siefert-Scintag automated powder diffractometer (PAD-II) using $\text{CuK}\alpha$ radiation. Patterns were step scanned ($0.03^\circ/5$ sec) and peaks located after data transfer to a larger computer. Peak positions were corrected by means of an internal calibrant (NBS-SRM #640 - Si powder) and cell parameters were found by indexing programs VISIND and TREOR¹³. Cell parameters were then least squared refined by program LSUCRE¹³.

Ionic conductivity measurements were performed on the samples in the form of sintered pellets by the complex impedance method.¹⁴ The pellets (8mm dia. and 2-3mm thick) were made at room temperature in an evacuative die at 10 tons/sq. inch and the pellets were sintered at 1150-1200°C for 24 hrs in a closed platinum crucible. These pressed and sintered pellets were polished with 400 grade emery paper and the faces were coated with conductive platinum paint (Englehard 6042) and heated at 600°C to

evaporate the solvent. For conductivity measurements the pellets were sandwiched between two platinum plates which were spring loaded to ensure good electrolyte/electrode contact. Impedance measurements were performed in the frequency region 5 Hz to 13 MHz using an HP 4192A impedance analyzer, and when required, to 108 MHz with an HP 4815 unit. Further details are given in ref. 15.

3. RESULTS

3.1 $\text{LiTi}_2(\text{PO}_4)_3$ - $\text{LiZr}_2(\text{PO}_4)_3$ Solid Solutions

$\text{LiZr}_2(\text{PO}_4)_3$ is known to be a poor conductor even though Li^+ is probably small enough to negotiate the bottlenecks leading to empty cavities.⁶ However, it was felt that it would be instructive to determine the effect on conductivity of decreasing the cavity size by progressive substitution of Ti^{4+} for Zr^{4+} in this compound. Cell parameters for members of the $\text{LiZr}_{2-x}\text{Ti}_x(\text{PO}_4)_3$ series are collected in Table 1.¹⁶ The X-ray pattern of $\text{LiZr}_2(\text{PO}_4)_3$, prepared at 1050-1100°C, was highly distorted from rhombohedral and possibly has a triclinic cell.⁶ Its cell parameters could not be deri-

TABLE 1
Unit Cell and Conductivity Data for $\text{LiZr}_{2-x}\text{Ti}_x(\text{PO}_4)_3$

x	Compound	a(Å)	c(Å)	$\sigma_{\text{RT}}(\text{ohm}^{-1}\text{cm}^{-1})$	$\sigma_{300^\circ\text{C}}(\text{ohm}^{-1}\text{cm}^{-1})$
0.0	$\text{LiZr}_2(\text{PO}_4)_3$	8.860 ^a	22.13 ^a	$<10^{-9}$	2.1×10^{-5}
0.1	$\text{LiZr}_{1.9}\text{Ti}_{0.1}(\text{PO}_4)_3$	8.845 ^b	22.057	3.2×10^{-8}	2.5×10^{-4}
0.2	$\text{LiZr}_{1.8}\text{Ti}_{0.2}(\text{PO}_4)_3$	8.828	21.978	5.6×10^{-8}	-
0.5	$\text{LiZr}_{1.5}\text{Ti}_{0.5}(\text{PO}_4)_3$	8.756	21.777	6.2×10^{-8}	5.2×10^{-4}
0.6	$\text{LiZr}_{1.4}\text{Ti}_{0.6}(\text{PO}_4)_3$	-	-	-	-
0.8	$\text{LiZr}_{1.2}\text{Ti}_{0.8}(\text{PO}_4)_3$	-	-	-	-
1.0	$\text{LiZrTi}(\text{PO}_4)_3$	-	-	4.0×10^{-8}	-
1.5	$\text{LiZr}_{0.5}\text{Ti}_{1.5}(\text{PO}_4)_3$	-	-	-	-
1.8	$\text{LiZr}_{0.2}\text{Ti}_{1.8}(\text{PO}_4)_3$	8.540	20.985	8.5×10^{-8}	7.4×10^{-3}
2.0	$\text{LiTi}_2(\text{PO}_4)_3$	8.504	20.840	7.9×10^{-8}	5.0×10^{-3}

^aExtrapolated values

^bErrors are ± 0.001 -2 in a and ± 0.003 -4 in c

ved from the X-ray pattern. However, since the cell parameters for the remaining members of the series were rhombohedral and fell on straight lines (Figure 1) values [for rhombohedral $\text{LiZr}_2(\text{PO}_4)_3$] were obtained by extrapolation.

An interesting variation in the nature of the X-ray patterns was observed as the titanium content of the solid increased. In the composition region $0.5 < x \leq 1.5$ the reflections broadened out to the point where accurate cell parameters could not be derived from the d spacings. This is illustrated in Table 1 by dashes in place of cell values and this composition region is indicated by dashed lines in Figure 1. Apparently the size disparity between the M^{4+} cations is sufficient to disorder the structure or cause lattice strain in this region of the solid solution. In the case of the ZrP_2O_7 - TiP_2O_7 system, a similar phenomenon was observed to the point that the solid solution series

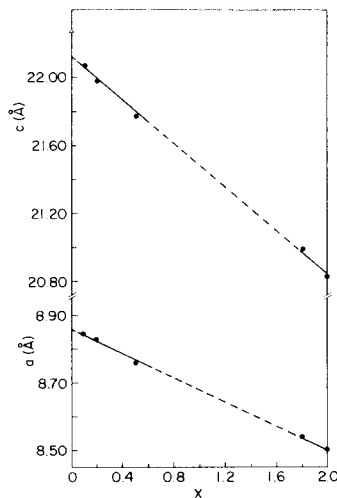


FIGURE 1
Cell parameters vs x for $\text{LiZr}_{2-x}\text{Ti}_x(\text{PO}_4)_3$.

TABLE 2
Unit Cell Parameters for Compositions in the Series $\text{Li}_{1+x}\text{Sc}_x\text{Ti}_{2-x}(\text{PO}_4)_3$

x	Composition	M/R ^a	$a(\text{\AA})$	$c(\text{\AA})$	Remarks
0.0	$\text{LiTi}_2(\text{PO}_4)_3$	R	8.504 ^b	20.840	
0.1	$\text{Li}_{1.1}\text{Sc}_{0.1}\text{Ti}_{1.9}(\text{PO}_4)_3$	R	8.522	20.889	
0.2	$\text{Li}_{1.2}\text{Sc}_{0.2}\text{Ti}_{1.8}(\text{PO}_4)_3$	R	8.536	20.922	
0.3	$\text{Li}_{1.3}\text{Sc}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$	R	8.559	20.964	
0.4	$\text{Li}_{1.4}\text{Sc}_{0.4}\text{Ti}_{1.6}(\text{PO}_4)_3$	R→M	8.570	20.995	Just becoming distorted.
0.5	$\text{Li}_{1.5}\text{Sc}_{0.5}\text{Ti}_{1.5}(\text{PO}_4)_3$	M			Monoclinic distortion, but could not be indexed.
0.8	$\text{Li}_{1.8}\text{Sc}_{0.8}\text{Ti}_{1.2}(\text{PO}_4)_3$	M			Monoclinic distortion, but could not be indexed.
1.0	$\text{Li}_2\text{ScTi}(\text{PO}_4)_3$				More complicated X-ray pattern could not be indexed.
	$\text{Li}_{1.2}\text{Sc}_{0.5}\text{Zr}_{0.2}\text{Ti}_{1.6}(\text{PO}_4)_3$	R	8.598	21.044	
	$\text{Li}_{1.5}\text{Sc}_{0.5}\text{Zr}_{0.5}\text{Ti}_{1.0}(\text{PO}_4)_3$	M			Very similar to $\text{Li}_{1.5}\text{Sc}_{0.5}\text{Ti}_{1.5}(\text{PO}_4)_3$, but reflections at lower θ .
	$\text{Li}_{1.2}\text{Sc}_{0.2}\text{Zr}_{1.6}\text{Ti}_{0.2}(\text{PO}_4)_3$	M			$a = 15.10\text{\AA}$, $b = 8.898\text{\AA}$, $c = 8.784\text{\AA}$, $\beta = 124.42^\circ$.

^aR = rhombohedral; M = monoclinic

^bErrors in $a_R \approx 0.001\text{\AA}$ and $b_R \approx 0.004\text{\AA}$

was not continuous.¹²

Measured conductivities for the different members of the solid solution are also shown in Table 1. The value given in the Table for $\text{LiZr}_2(\text{PO}_4)_3$ is for the triclinic (or possibly monoclinic) phase. It is seen that $\text{LiTi}_2(\text{PO}_4)_3$ exhibits a conductivity about two orders of magnitude greater than the zirconium compound. However, there is not a smooth transition from one end member to the other. In the region of high disorder the powders were difficult to sinter and consequently the compacts were of lower density. Thus, these values are subject to greater error. If we disregard the value for $\text{LiZrTi}(\text{PO}_4)_3$, the conductivity appears to increase with a decrease in cell parameters until a maximum in the conductivity is achieved for $\text{LiZr}_{0.2}\text{Ti}_{1.8}(\text{PO}_4)_3$.

3.2 Sc^{3+} Substitution in $\text{LiTi}_2\text{P}_3\text{O}_{12}$

In order to improve the conductivity, Sc^{3+} was added as a partial replacement for Ti^{4+} according to the formula $\text{Li}_{1+x}\text{Sc}_x\text{Ti}_{2-x}(\text{PO}_4)_3$. Unit cell data for compositions with values of x between 0 and 1 are shown in Table 2 and plotted in Figure 2. Since Sc^{3+} is slightly larger than Ti^{4+} the effect of the substitution is to increase the cell parameters in a linear fashion. However, when $x = 0.5$ the powder lines split in a manner indicative of formation of a lattice of lower symmetry, probably monoclinic. However, we could not index the pattern. At a value of $x = 1.0$, further splitting of the lines occurs and the crystal system could not be identified.

Conductivity data for the scandium containing solids are shown in Table 3. A dramatic increase in the magnitude of the conductivity appears with respect to the mixed titanium-zirconium composition. The differences in cell volumes for the scandium containing solids versus the Ti-Zr mixed compositions are very minor. Thus, the bulk of the conductivity change must

result from the increase in Li^+ content. It is further seen that the conductivity passes through a maximum at $x = 0.3$. This may represent the point at which the cavities become very large relative to the size of the lithium ion resulting in stronger binding of the cation to the lattice oxygens.

Arrhenius plots of $\log \sigma$ versus the reciprocal of the absolute temperature are shown in Figure 3. It is seen that the curves divide into two straight lines. The slopes below 100°C give values of $E_a = 0.1$ eV for $x = 0.2$ and 0.3 , respectively. At the higher temperature range the activation energies are about 4 times higher.

Data is also presented at the lower end of Table 2 and 3 for the substitutions of Sc^{3+} in the mixed titanium-zirconium compositions. Comparison of the cell parameters for $\text{Li}_{1.2}\text{Sc}_{0.2}\text{Zr}_{0.2}\text{Ti}_{1.6}(\text{PO}_4)_3$ with the values in Table 1 shows a normal increase in the cell

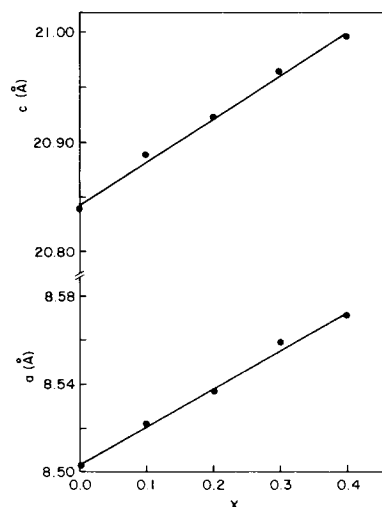


FIGURE 2
Cell parameters as a function of Sc^{3+}
content for $\text{LiTi}_{2-x}(\text{PO}_4)_3$.

TABLE 3
 Conductivity Data for $\text{Li}_{1+x}\text{Sc}_x\text{Ti}_{2-x}(\text{PO}_4)_3$

Compound	$\sigma_{\text{RT}} (\text{ohm}^{-1}\text{cm}^{-1})$	$\sigma_{300^\circ\text{C}} (\text{ohm}^{-1}\text{cm}^{-1})$	$E_a (\text{eV})$
$\text{Li}_{1.1}(\text{Sc}_{0.1}\text{Ti}_{1.9})(\text{PO}_4)_3$	1.58×10^{-4}	1.4×10^{-2}	-
$\text{Li}_{1.2}(\text{Sc}_{0.2}\text{Ti}_{1.8})(\text{PO}_4)_3$	2.46×10^{-4}	2.6×10^{-2}	0.36
$\text{Li}_{1.3}(\text{Sc}_{0.3}\text{Ti}_{1.7})(\text{PO}_4)_3$	4.07×10^{-4}	6.3×10^{-2}	0.40
$\text{Li}_{1.4}(\text{Sc}_{0.4}\text{Ti}_{1.6})(\text{PO}_4)_3$	2.20×10^{-4}	4.9×10^{-2}	-
$\text{Li}_{1.8}(\text{Sc}_{0.8}\text{Ti}_{1.2})(\text{PO}_4)_3$	8.80×10^{-4}	1.0×10^{-2}	-
$\text{Li}_{1.2}\text{Sc}_{0.2}\text{Zr}_{0.2}\text{Ti}_{1.6}\text{P}_{0.12}\text{O}_{12}$	1.20×10^{-4}	1.7×10^{-2}	0.33
$\text{Li}_{1.2}\text{Sc}_{0.2}\text{Zr}_{1.5}\text{Ti}_{0.2}\text{P}_{0.12}\text{O}_{12}$	3.10×10^{-7}	7.9×10^{-4}	0.43

^aTemperature range 100–300°C

dimensions. However, when the scandium level was increased, a transition, probably to a monoclinic lattice, occurs. The X-ray pattern resembles that for $\text{Li}_{1.5}\text{Sc}_{0.5}\text{Ti}_{1.5}(\text{PO}_4)_3$ with the reflection shifted slightly to a lower angle indicative of larger cell parameters. At the high Zr end of the series,

$\text{Li}_{1.2}\text{Sc}_{0.2}\text{Zr}_{1.6}\text{Ti}_{0.2}(\text{PO}_4)_3$, the powder pattern could be indexed with a monoclinic cell. Conductivity data for two of the Sc-Ti-Zr compositions is shown in Table 3 and Figure 4.

3.3 In^{3+} Substitution in $\text{LiHf}_2(\text{PO}_4)_3$

Taylor et al.⁶ have reported that $\text{LiHf}_2(\text{PO}_4)_3$ crystallizes with a rhombohedral structure similar to $\text{NaZr}_2(\text{PO}_4)_3$. We attempted to improve the conductivity of $\text{LiHf}_2(\text{PO}_4)_3$ by substituting Hf with a slightly larger cation, In^{3+} . Cell parameters for $\text{Li}_{1+x}\text{In}_x\text{Hf}_{2-x}(\text{PO}_4)_3$ compositions are given in Table 4. In this case, $\text{R}\bar{3}\text{c}$ solid solutions exist up to $x = 3$. For higher values of x , the X-ray patterns were complex due to the lowering of symmetry. Conductivity data for the above compositions are also given in Table 4 and Figure 5. As observed by Taylor et al., the conductivity of $\text{LiHf}_2(\text{PO}_4)_3$ is higher than that of the corresponding Zr compound (Table 1). Substitution of Hf by In^{3+} increased the Li ion conductivity slightly within the $\text{R}\bar{3}\text{c}$ solid solutions.

4. DISCUSSION

The crystal structure of triphosphates of the type $\text{A}^{\text{I}}\text{M}_2^{\text{IV}}(\text{PO}_4)_3$ consists of a three dimensional network of phosphate tetrahedra and MO_6 octahedra sharing corners.^{2,3,8} This arrangement forms $\text{Zr}_2(\text{PO}_4)_3^-$ groups and each group is linked to others by PO_4 tetrahedra. Two adjacent such groups along the C_r axis, together with linking phosphates, form a cavity in which

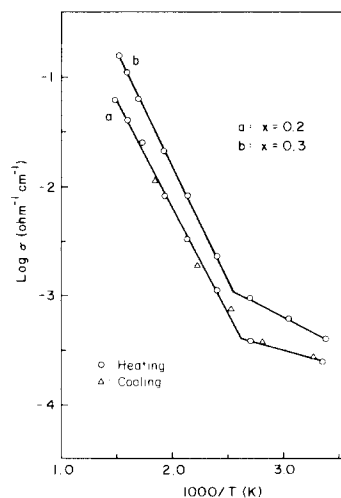


FIGURE 3
 Specific conductivity as a function of $1/T$
 for $\text{Li}_{1+x}\text{Sc}_x\text{Ti}_{2-x}(\text{PO}_4)_3$

TABLE 4
Unit Cell and Conductivity Data for $\text{Li}_{1+x}\text{In}_x\text{Hf}_{2-x}(\text{PO}_4)_3$

x	a(Å)	c(Å)	σ_{RT} ($\text{ohm}^{-1}\text{cm}^{-1}$)	$\sigma_{300^\circ\text{C}}$ ($\text{ohm}^{-1}\text{cm}^{-1}$)	E_a (eV)
0.0	8.809	21.980	3×10^{-6}	4.5×10^{-3}	0.44
0.1	-	-	1×10^{-6}	8.5×10^{-3}	0.43
0.2	8.815	21.954	4×10^{-6}	8.0×10^{-2}	0.44
0.3	8.820	21.933	1×10^{-5}	3.2×10^{-2}	0.45
0.4	(Distorted from $\text{R}\bar{3}\text{c}$)		4×10^{-6}	5.0×10^{-3}	0.48

the A ion resides. All of these cavities are filled and are termed type I. There are three other cavities, or A cation sites, per formula which lie in layers above and below. Adjacent layers are shifted by $1/2$ along the C_r axis. The new cavities (type II) which arise from this positioning of the layers have 8-10 oxygen atoms which coordinate Na^+ compared to six in a trigonal prism arrangement in the type I cavities which coordinate Na^+ . In the NASICON solid solution, $\text{Na}_{1+x}\text{Zr}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$, sodium ions fill the type II cavities as their numbers increase with increasing silicon content. At the same time, the passageways connecting the cavities increase in size allowing freer movement of sodium ions with consequent rise in conductivity. In contrast, the very low conductivity of $\text{NaZr}_2(\text{PO}_4)_3$ results from the narrowness of the passageways leading from type I to type II cavities which severely restrict Na^+ mobility. Another factor working against high mobility of the conducting ions is their tight binding to the framework oxygens. This factor must become very important in the Li^+ compounds as discussed below.

A very large increase in conductivity takes place upon substitution of Ti^{4+} for Zr^{4+} in the framework and a further increase takes place when relatively small amounts of Sc^{3+} are substituted for Ti^{4+} in $\text{LiTi}_2(\text{PO}_4)_3$. A maximum in the conductivity was achieved for $\text{Li}_{1.3}\text{Sc}_{0.3}\text{Ti}_{1.7}\text{P}_3\text{O}_{12}$ and shows that there is a critical interplay of several factors respon-

sible for increased conductivity. We may enumerate these in keeping with previous deductions³ as (1) The size of the cavities in which the Li^+ resides; (2) The size of the bottlenecks to diffusion; (3) The strength of the Li-lattice oxygen bonds. Of course, increasing the number of conductors while leaving sufficient cavities empty must also be considered. Some of these factors are clearly illustrated by the present study. Most striking is the increase in conduc-

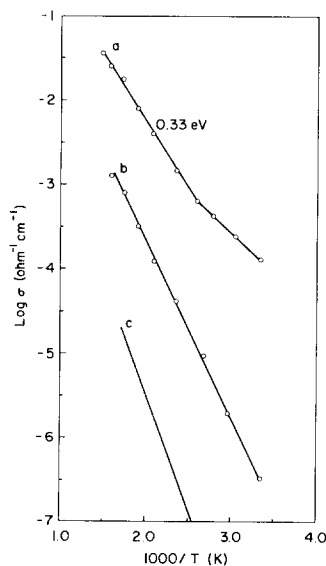


FIGURE 4
Specific conductivity as a function of $1/T$ for
a) $\text{Li}_{1.2}\text{Sc}_{0.2}\text{Ti}_{1.6}\text{Zr}_{0.2}(\text{PO}_4)_3$
b) $\text{Li}_{1.2}\text{Sc}_{0.2}\text{Ti}_{0.2}\text{Zr}_{1.6}(\text{PO}_4)_3$
c) $\text{LiZr}_2(\text{PO}_4)_3$

tivity when the smaller titanium ion is substituted for Zr^{4+} . This change undoubtedly decreases the size of the cavities as evidenced by the large decrease in cell parameters ($\sim 0.3\text{\AA}$ in both a and c). The closer proximity of Li^+ to Ti^{4+} would result in a greater repulsion from the Ti^{4+} ions surrounding the cavity and tend to produce weaker Li-O bonds. However, the optimum in conductivity was achieved for $\text{LiZr}_{0.2}\text{Ti}_{1.8}\text{P}_3\text{O}_{12}$ where the cell parameters are slightly larger, which may be connected with slight changes in bottleneck size also.

During the course of this work we became aware of a similar study⁷ in which In^{3+} substitutions for Ti^{4+} were carried out. In this case the maximum was achieved at 0.4 mole In. The ionic radii of In^{3+} and Sc^{3+} are almost identical but this difference in composition may be reflecting some small differences in electron configuration of the two ions.

The conductivity of pure $\text{LiHf}_2(\text{PO}_4)_3$ is two orders of magnitude larger than that of $\text{LiZr}_2(\text{PO}_4)_3$ in accordance with results obtained by Taylor et al.⁶ However, increase in the

number of Li conductors by replacement of Hf by In did not result in a drastic increase in Li^+ ion conductivity. If one compares the unit cell and conductivity data for $\text{Li}_{1.3}\text{Sc}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ with $\text{Li}_{1.3}\text{In}_{0.3}\text{Hf}_{1.7}(\text{PO}_4)_3$ the former has a lower unit cell volume and higher conductivity (Tables 2, 3 and 4). This indicates that the bottleneck size of the conducting path plays an important role among Li ion conductors based on the $\text{LiM}_2(\text{PO}_4)_3$ type structure.

From the results reported here it would appear that the course to follow to achieve even higher Li^+ conductivity is proper substitution into the $\text{LiTi}_2\text{P}_3\text{O}_{12}$ lattice so as to increase the Li^+ content while simultaneously adjusting cavity and bottleneck size to allow free movement of the ions. In this connection we have achieved a composition of $\text{Li}_{1.6}\text{Ga}_{0.6}\text{Ti}_{1.4}(\text{PO}_4)_3$ with $R\bar{3}c$ structure. Conductivity measurements are in progress

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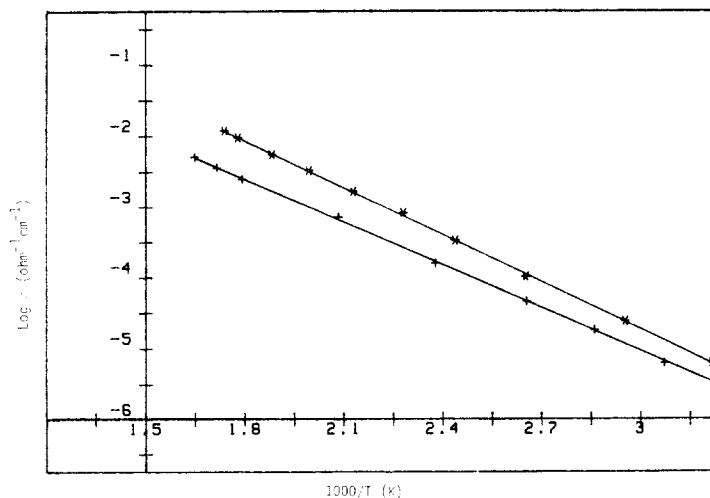


FIGURE 5
Log σ vs $1/T$ plots for $\text{Li}_{1+x}\text{In}_x\text{Hf}_{2-x}(\text{PO}_4)_3$ compounds
* : $x = 0.2$; o : $x = 0.0$

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